ALLIANCE UNIVERSITY

Private University established in Karnataka State by Act No.34 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi

Alliance School of Advanced Computing

Department of Computer Science and Engineering

Class Assignment-1

Course Code: 5CS1025

Course Title: Artificial Intelligence

Semester: 04

Class : AIML

Name: MOUNA.R

Reg no: 2023BCSE07AED408

GitHub: https://github.com/Mouna-R408/Mouna-R408

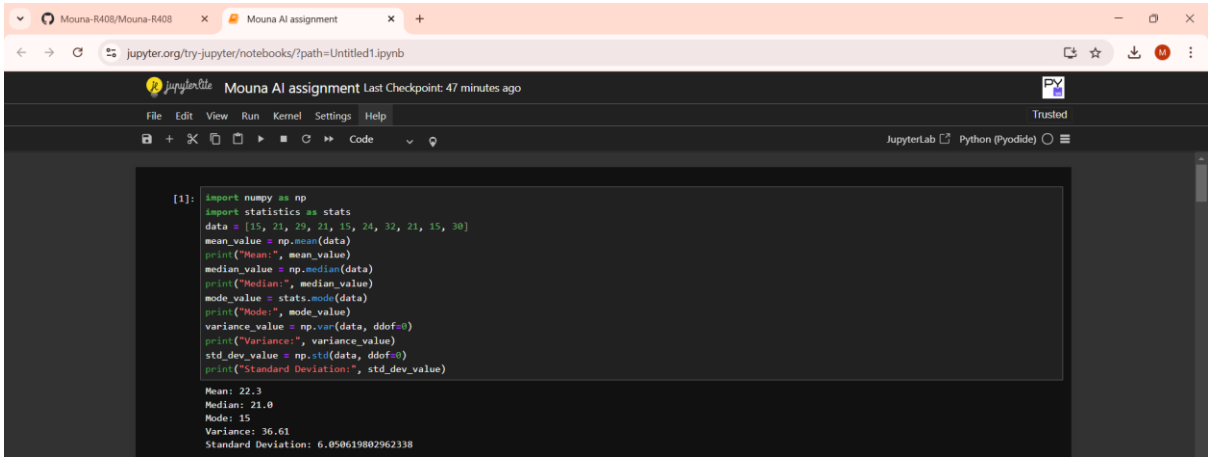
2024-25

1.Imagine you are tasked with designing a humanoid robot to assist in a home or office environment. The robot must be capable of interacting with people by talking and listening, walking to different locations, seeing and recognizing objects, and learning from its surroundings to adapt its behavior. What technologies, tools, and frameworks would you need to build such a robot? Give as flow chart

```
+-------------------------------------------------------------+
|                    HUMANOID ROBOT SYSTEM                    |
+-------------------------------------------------------------+
          |
          v
   +---------------------+
   |  Hardware Layer     |
   +---------------------+
   |  - Motors & Actuators (for movement) |
   |  - Sensors (Camera, LIDAR, IMU) |
   |  - Microcontroller/Embedded System (e.g. Raspberry Pi, NVIDIA Jetson) |
   +---------------------+
          |
          v
   +------------------------+
   |  Perception & Sensing Layer  |
   +------------------------+
   |  - Computer Vision (OpenCV, TensorFlow, PyTorch) |
   |  - Object Recognition (YOLO, Faster R-CNN) |
   |  - Audio Processing (Speech-to-Text, NLP) |
   |  - Environment Sensing (LIDAR, Depth Sensors) |
   |  - Localization & Mapping (SLAM, Visual Odometry) |
   +------------------------+
          |
          v
   +--------------------------+
   |  Decision & Control Layer |
   +--------------------------+
   |  - Path Planning (A*, Dijkstra, RRT) |
   |  - Reinforcement Learning (RL) |
   |  - Behavior Modeling (Finite State Machine, Reactive Planning) |
   +--------------------------+
          |
          v
   +------------------------+
   |  Interaction & Learning Layer |
   +------------------------+
   |  - Speech Synthesis & Recognition (Google Speech API, Microsoft Cognitive Services) |
   |  - Natural Language Processing (Dialogflow, Rasa NLU) |
```

```
|  - Emotion Detection (Facial Recognition, Sentiment Analysis) |
|  - Machine Learning Frameworks (TensorFlow, PyTorch, Keras) |
+-----------------------+
        |
        v
+----------------------+
|  Communication Layer |
+----------------------+
|  - Cloud Integration (AWS, Azure, Google Cloud) |
|  - Wireless Communication (Wi-Fi, Bluetooth, Zigbee) |
|  - IoT Integration (Sensors, Devices) |
+----------------------+
        |
        v
+----------------------+
|  User Interface & API |
+----------------------+
|  - GUI/UX for robot control |
|  - REST APIs, WebSockets |
|  - Mobile App/Cloud Dashboards |
```

2. Calculate and interpret mean, median, mode, variance and standard deviation for a given dataset. Data =[15,21,29,21,15,24,32,21,15,30]



3. You are analyzing a dataset that captures the daily performance and activity of a humanoid robot in a simulated environment. The dataset link robot_dataset(robot_dataset)_1.csv includes the following attributes

Interaction_Count: Number of conversations the robot had daily.
Steps_Walked: Total steps taken each day.
Objects_Recognized: Number of objects successfully identified by the robot.
Learning_Sessions: Number of learning tasks completed.
Energy_Consumption (kWh): Daily energy usage of robots.

Mouna-R408/Mouna-R408 | Mouna AI assignment

jupyter.org/try-jupyter/notebooks/?path=Untitled1.ipynb

Mouna AI assignment Last Checkpoint: 47 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python (Pyodide)

[2]:
```
import pandas as pd
df=pd.read_csv("robotdataset.csv")
df
```

[2]:

	Robot_ID	Task_Type	Component_ID	Sensor_Type	Sensor_Data	Processing_Time (s)	Accuracy (%)	Environmental_Status	Energy_Consumption (kWh)	Human_Intervention_Neede
0	RBT_001	Inspection	CMP_460	LIDAR	1 (obstacle detected)	67.0	90.4	Stable	2.2	N
1	RBT_002	Assembly	CMP_252	Thermal	85.3 (°C)	71.2	98.1	Stable	2.7	Ye
2	RBT_003	Inspection	CMP_248	Thermal	92% (visual fit)	49.2	95.3	Unstable	2.4	N
3	RBT_004	Welding	CMP_433	Camera	98% (defect-free)	74.5	90.2	Stable	2.4	Ye
4	RBT_005	Assembly	CMP_992	Camera	92% (visual fit)	64.5	97.2	Unstable	1.8	N
5	RBT_006	Painting	CMP_309	Thermal	1 (obstacle detected)	51.6	93.4	Unstable	1.5	N
6	RBT_007	Assembly	CMP_637	Thermal	1 (obstacle detected)	50.9	95.5	Stable	1.8	Ye
7	RBT_008	Inspection	CMP_571	Camera	85.3 (°C)	61.7	90.2	Unstable	2.3	Ye
8	RBT_009	Assembly	CMP_630	Camera	0 (clear path)	49.4	92.9	Stable	1.7	Ye
9	RBT_010	Painting	CMP_617	LIDAR	0 (no obstacle)	65.1	94.1	Stable	2.9	N
10	RBT_011	Assembly	CMP_567	Camera	0 (no obstacle)	48.5	96.1	Unstable	2.6	Ye

Perform Basic Statistical Operations:

1)What is the average (mean) number of conversations the robot has daily?

Mouna-R408/Mouna-R408 | Mouna AI assignment

jupyter.org/try-jupyter/notebooks/?path=Untitled1.ipynb

Mouna AI assignment Last Checkpoint: 47 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python (Pyodide)

9	RBT_010	Painting	CMP_617	LIDAR	0 (no obstacle)	65.1	94.1	Stable	2.9	N
10	RBT_011	Assembly	CMP_567	Camera	0 (no obstacle)	48.5	96.1	Unstable	2.6	Ye
11	RBT_012	Painting	CMP_268	LIDAR	75.8 (°C)	44.8	99.6	Stable	2.2	N
12	RBT_013	Inspection	CMP_631	Thermal	98% (defect-free)	53.2	93.8	Unstable	2.0	N
13	RBT_014	Inspection	CMP_115	Camera	82.4 (°C)	41.7	92.0	Stable	2.2	Ye
14	RBT_015	Inspection	CMP_932	LIDAR + Camera	92% (visual fit)	50.0	91.3	Stable	2.8	Ye
15	RBT_016	Assembly	CMP_424	LIDAR + Camera	92% (visual fit)	57.0	98.1	Unstable	1.6	Ye
16	RBT_017	Inspection	CMP_201	Camera	0 (no obstacle)	78.1	98.9	Stable	2.9	Ye
17	RBT_018	Painting	CMP_262	LIDAR + Camera	92% (visual fit)	49.3	92.0	Unstable	1.8	Ye

[3]:
```
mean_specific = df['Interaction_Count'].mean()
print("\nMean of the specific column 'Interaction_Count':", mean_specific)
```

```
5.0 of the specific column 'Interaction_Count':
```

2) Find the total steps walked by the robot over a given period.

Mouna-R408/Mouna-R408 | Mouna AI assignment

jupyter.org/try-jupyter/notebooks/?path=Untitled1.ipynb

Mouna AI assignment Last Checkpoint: 47 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python (Pyodide)

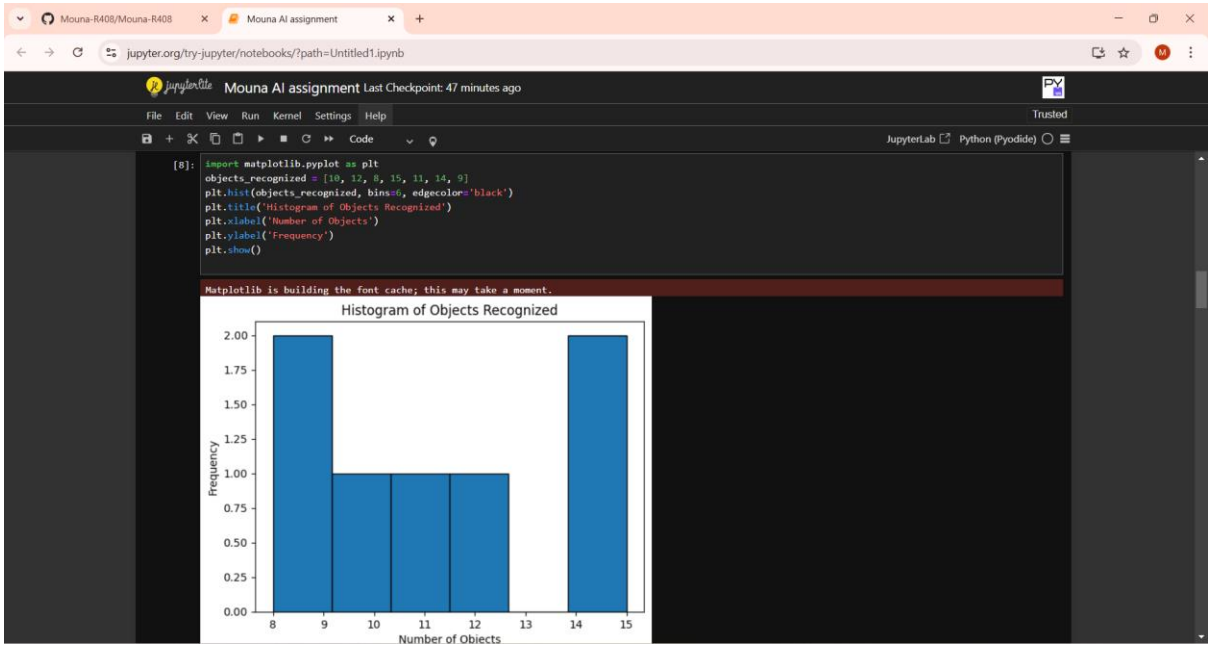
[4]:
```
total = df['Steps_Walked'].sum()
print(f"The total of the column is: {total}")
```

```
The total of the column is: 424
```

3) Determine the maximum and minimum energy consumption in the dataset.

4) Calculate the correlation between the number of steps walked and energy consumption.

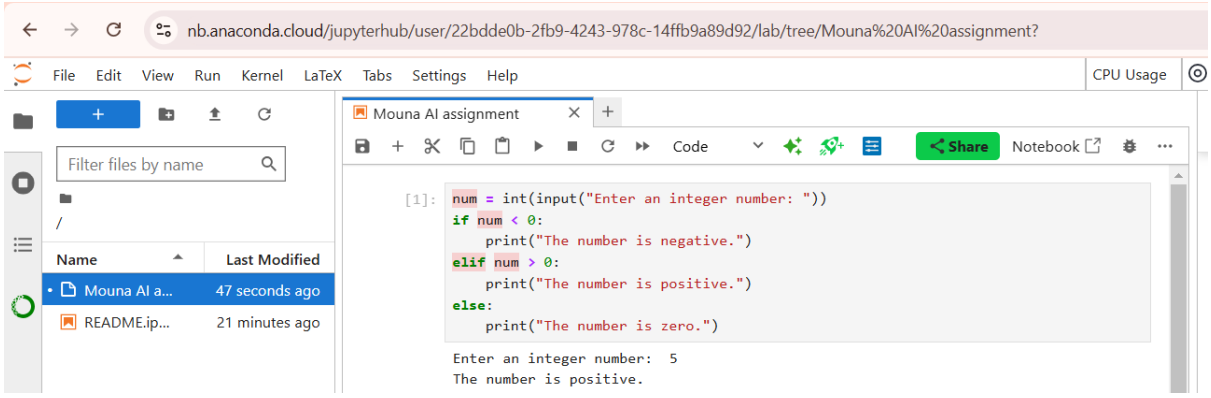
5) Analyze the distribution of objects recognized daily (e.g., histogram or box plot).



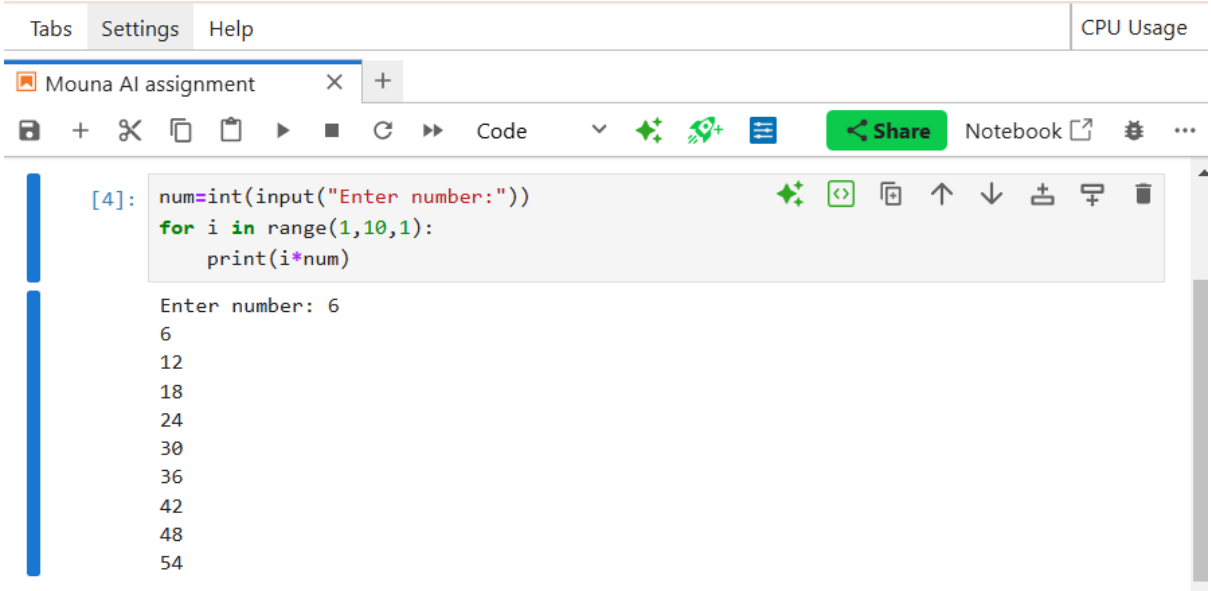
6) What is the variance in the number of learning sessions completed?

4. Write a Python program that declares variables of different data types (e.g., string, integer, float, and boolean). Output the variables in a sentence format using print() and f-strings.

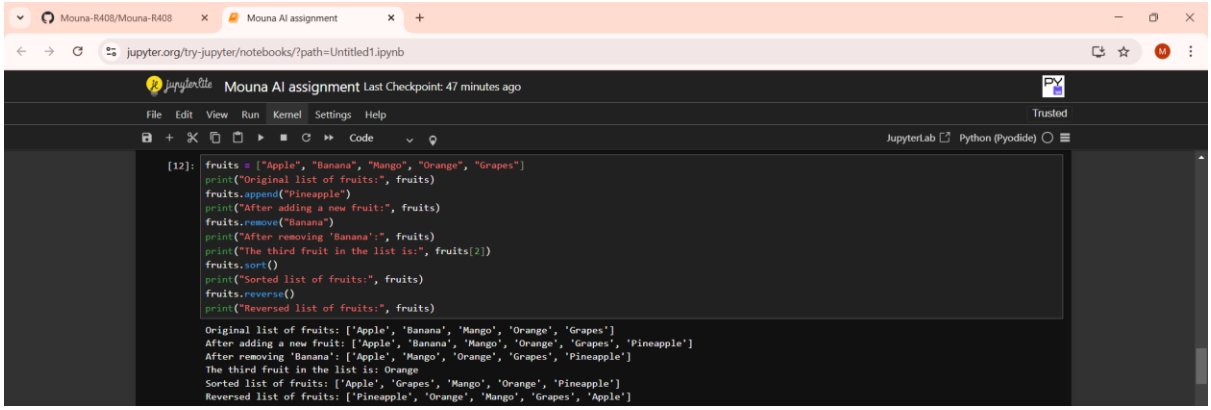
5. Write a Python program that takes an integer input and checks whether the number is positive, negative, or zero using conditional statements (if-else).



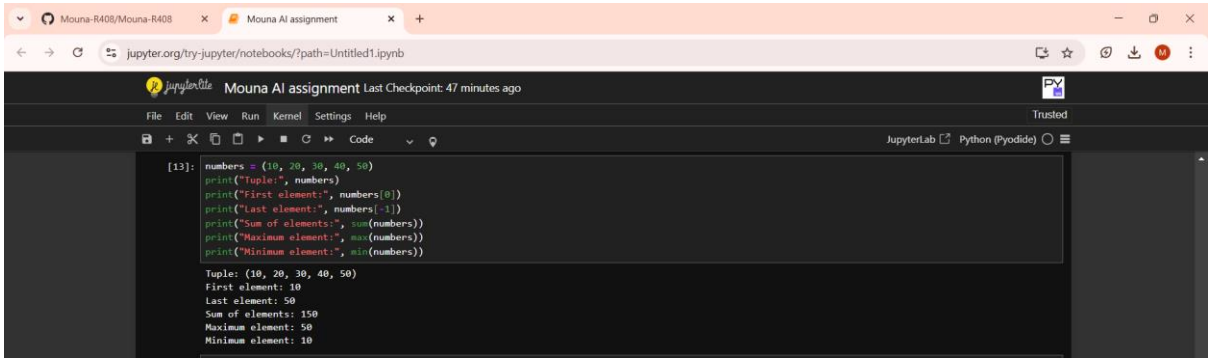
6. Write a Python program that takes a number as input and prints the multiplication table for that number (from 1 to 10).



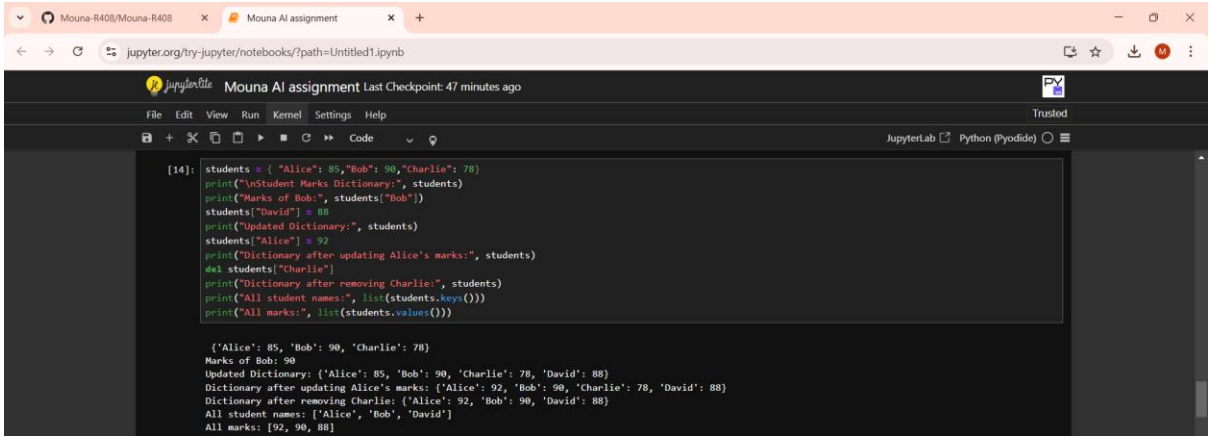
7. Create a Python list that contains the names of 5 different fruits. Perform the given operations on the list.



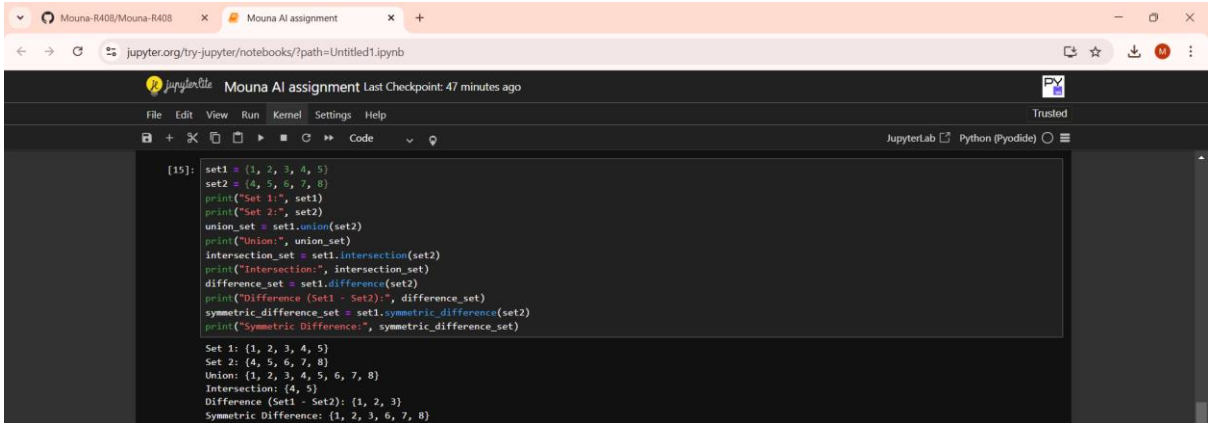
8. Write a Python program that creates a tuple containing 5 numbers. Perform the given operations on the tuple.



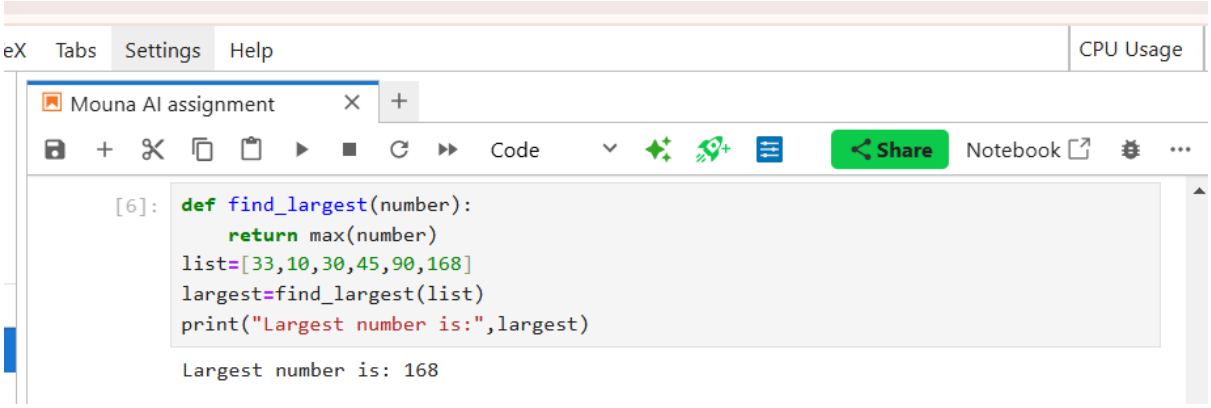
9. Create a dictionary that stores the names of 3 students as keys and their marks in mathematics as values. Perform the given operations



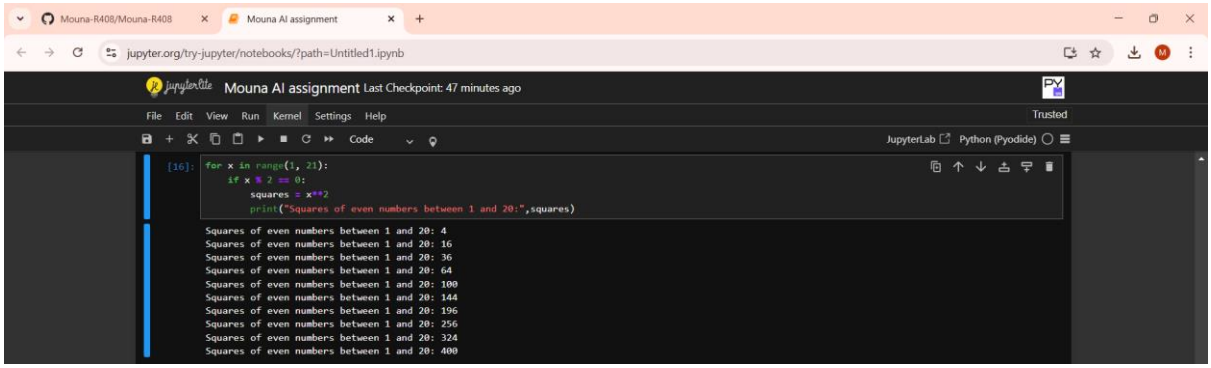
10. Create two sets of integers. Perform the given set operations



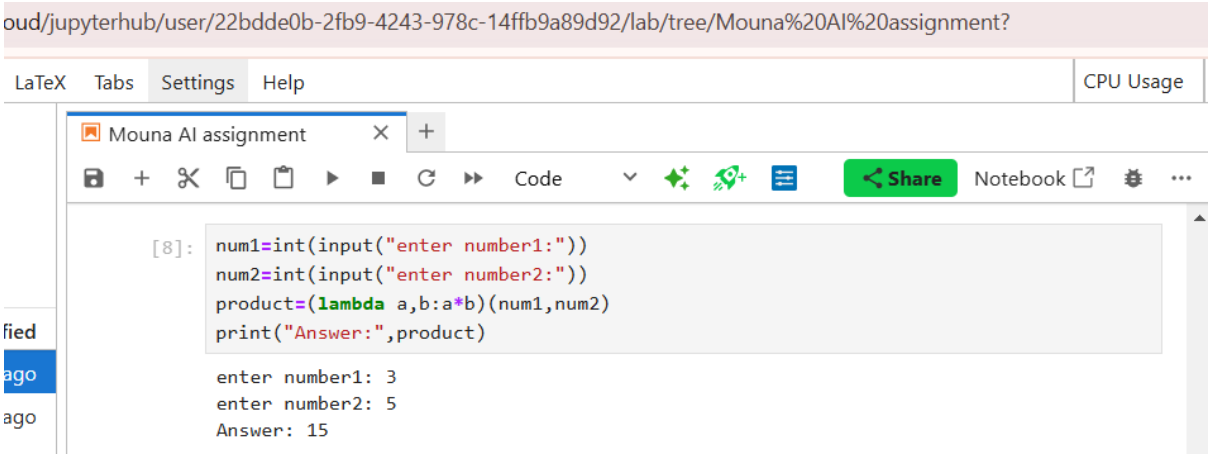
11. Write a Python function called find_largest() that takes a list of numbers as input and returns the largest number from the list. Test the function with a sample list.



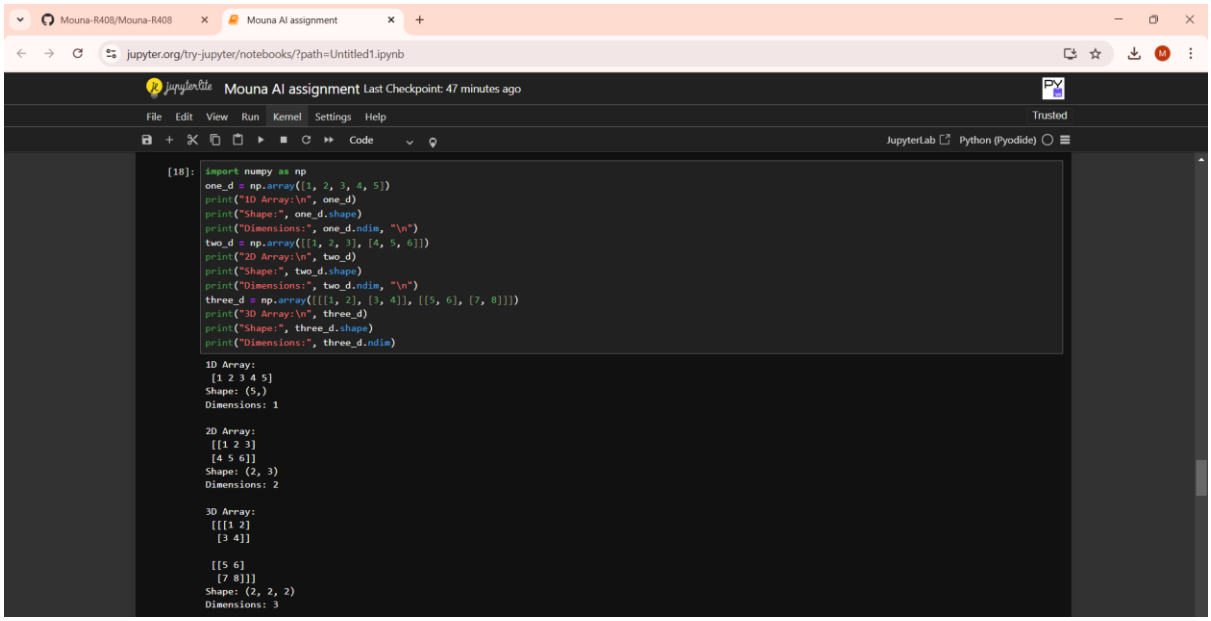
12. Use list comprehension to create a list of squares of all even numbers between 1 and 20



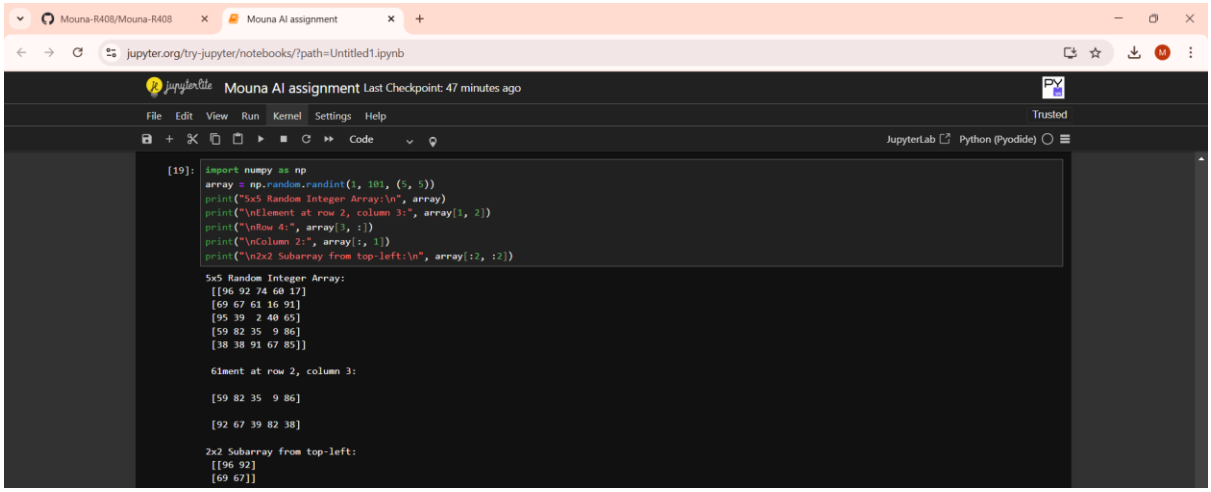
13. Write a Python script that uses a lambda function to calculate the product of two numbers provided by the user.



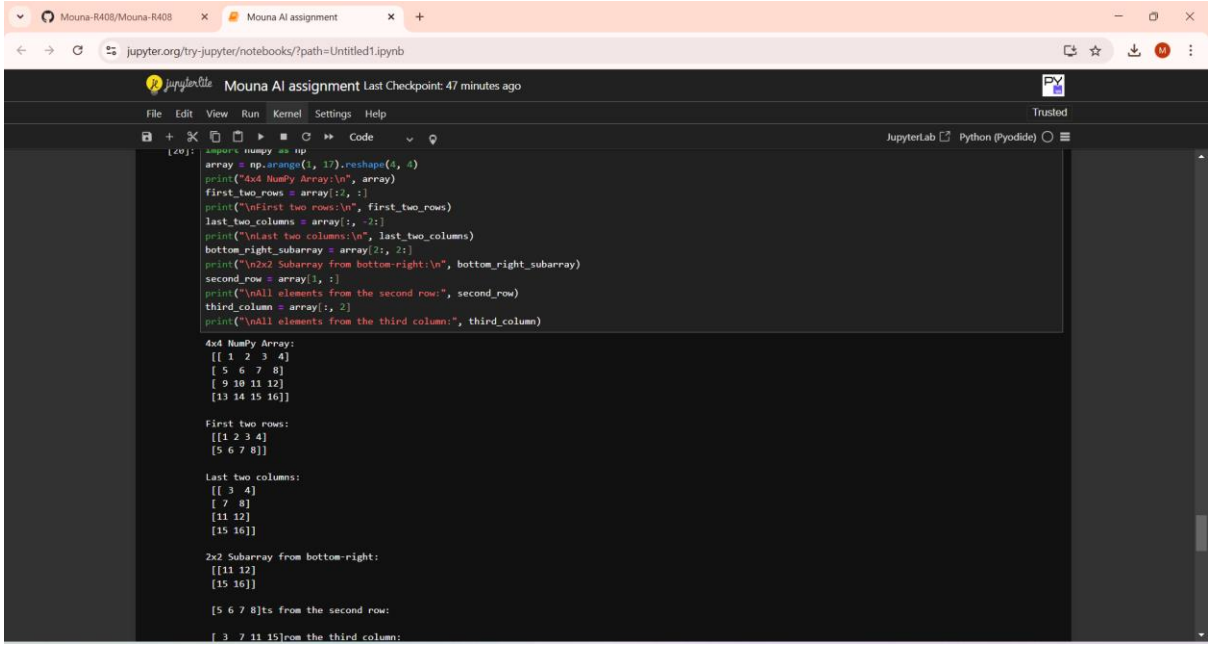
14. Write a Python program to create a one-dimensional, two-dimensional, and three-dimensional NumPy array. Print the shape and dimensions of each array.



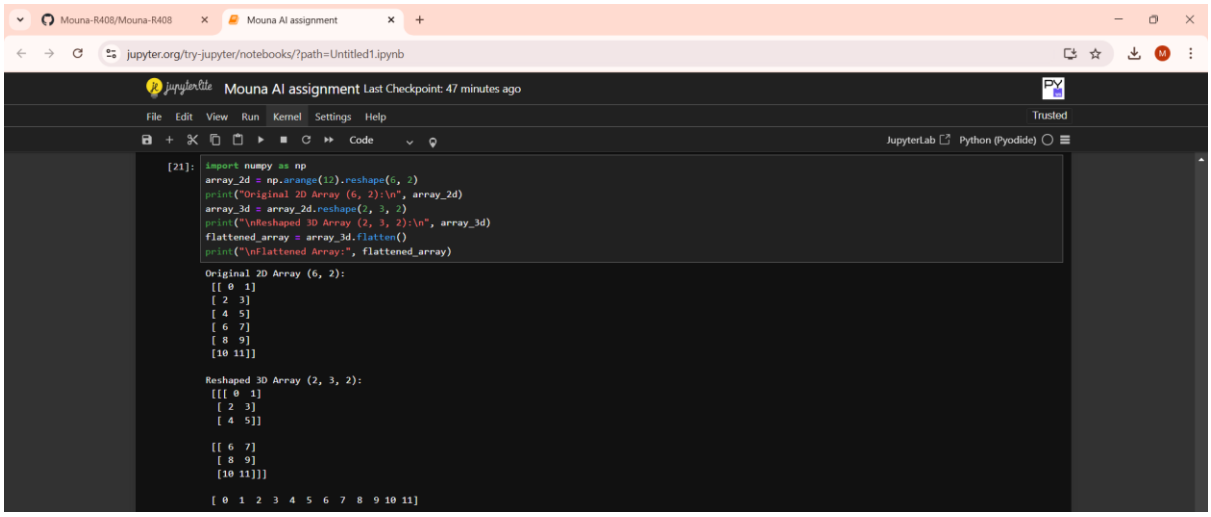
15. Write a Python program to create a 5x5 NumPy array of random integers and Perform array indexing as given.



16. create a NumPy array of shape (4, 4) containing numbers from 1 to 16. Use slicing to extract for the given conditions



17. Write a Python program that creates a 2D array of shape (6, 2) using np.arange() and then reshapes it into a 3D array of shape (2, 3, 2). Flatten the reshaped array and print the result.



18. Write a Python program to demonstrate broadcasting. Create an array of shape (3, 3) and add a onedimensional array of shape (1, 3) to it using broadcasting.

19. Create two NumPy arrays of the same shape, A and B. Perform the following arithmetic operations: Element-wise addition. Element-wise subtraction. Element-wise multiplication. Element-wise division.

eX Tabs Settings Help CPU Usage

Mouna AI assignment

Code Share Notebook

[18]:
```
import numpy as np
a=np.array([1,2,3])
b=np.array([2,3,4])
addition=a+b
subtraction=a-b
multiplication=a*b
division=a/b
print(addition)
print(multiplication)
print(division)
print(subtraction)
```

```
[3 5 7]
[ 2  6 12]
[0.5        0.66666667 0.75      ]
[-1 -1 -1]
```

20. Create a Pandas DataFrame with the given Name and marks of 3 courses: Add a new column named 'Total' that represents the sum of all the courses. Add 'Grade' based on the values of the 'Total'. Print the updated DataFrame with the new 'Total' and 'Grade' column.

upyterhub/user/22bdde0b-2fb9-4243-978c-14ffb9a89d92/lab/tree/Mouna%20AI%20assignment?

X Tabs Settings Help CPU Usage

Mouna AI assignment

Code Share Notebook

[16]:
```
import pandas as pd
data={'Name':["Mouna","megha","anu"],
      'course1':[88,90,92],
      'course2':[78,99,90],
      'course3':[99,100,85]}
df=pd.DataFrame(data)
df['Total']=df[['course1','course2','course3']].sum(axis=1)
df['Grade']=df['Total'].apply(lambda a:'A' if a>=270 else ('B' if a>=240 else 'c'))
print(df)
```

```
    Name  course1  course2  course3  Total Grade
0  Mouna       88       78       99    265     B
1  megha       90       99      100    289     A
2    anu       92       90       85    267     B
```